Cyberjaya 5G Future Talent Innovation Challenge 2026 · Track 2: 5G Connected & Safe Last-Mile Mobility

MATA KEDUA: Roadside Views for the Remote Assistance of Driverless Delivery Vans in Cyberjaya

A desk-based design study

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Abstract. Since 20 October 2026, a fully driverless delivery van has run on Cyberjaya's 22 km MyAV route, supervised from a Remote Operations Centre that, by public accounts, watches the van's own video and telemetry. When such a vehicle meets a situation it cannot resolve, it stops and asks a remote person for help, and that person sees only what the vehicle's cameras see. A double-parked car at a school gate can hide a child from all of them. This study asks whether an on-request roadside view at known hotspots, carried over 5G, could close that gap in a Malaysian setting, and what such a system would need. We reviewed about 70 public sources (about 35 read in full) on remote-assistance practice, roadside support for remote monitoring, Malaysian mobile-network measurements and Malaysian road-safety evidence, and we built an interactive prototype of the resulting design. The review finds that "advise, never steer" assistance is established practice; that roadside views for remote monitors already run in Japan but have no published Malaysian precedent; and that the binding network constraint is upload capacity at busy moments rather than latency. Four HD camera views need roughly 16–32 Mbps of upload, against average 5G uploads of about 31–33 Mbps and all-network averages of 15–17 Mbps for Malaysia's largest operators (April–June 2026). We propose MATA KEDUA, a design that adds a roadside view, an interlock that keeps "Go" locked while that view shows people in the path, and a Malaysian edge-case library. No primary data were collected; we specify a Phase 1 protocol to test the claims on the route.

Keywords: remote assistance; automated vehicles; 5G uplink; roadside infrastructure; pedestrian safety; school zones; Malaysia

1. Introduction

On 20 October 2026, Autonomous Logistic Solutions (ALS) began a six-month trial of a Level 4 delivery van, the RoboVan, on Cyberjaya's 22 km MyAV route. It carries no driver and no passengers (NST, 2026a). The route "covers business districts, educational institutions, MRT stations, commercial centres, intersections and pedestrian crossings" (Business Today, 2026). The van is supervised from a Remote Operations Centre (ROC) that ALS launched with Sidec and Cyberview on 31 July 2026. Its operators monitor "live video feeds, vehicle telemetry, route status and system diagnostics" and coordinate with ground teams when needed (Business Today, 2026). No public source says the centre receives views from roadside cameras.

Driverless vehicles rarely fail dramatically. More often they hesitate: at a construction layout, a closed lane, an officer directing traffic, or an object they cannot classify (Waymo, 2024). The industry's answer is remote assistance. The vehicle stops safely and asks a question, and a person answers it. The person decides from the vehicle's own sensors. In most situations that is enough. At a school gate during drop-off, with cars double-parked, it may not be: the van's cameras see the back of an SUV, and nobody in the loop can see the child between the cars.

That scene is not hypothetical. On 23 January 2026, an unoccupied Waymo travelling at 17 mph struck a nine-year-old who crossed midblock from between queued and double-parked cars near a Santa Monica primary school during drop-off. The child suffered minor injuries. The National Highway Traffic Safety Administration opened a preliminary evaluation into whether the car "exercised appropriate caution" near the school (NHTSA, 2026), and the National Transportation Safety Board noted video from a school surveillance camera (NTSB, 2026b). The car did not ask for remote assistance before the impact, so the case shows the hazard, not a failure of assistance. It does show that a camera at the school saw what the car's cameras could not.

The Malaysian stakes are high. Government figures put the toll at an average of 67 deaths a year in school zones, counting all road users (WHO, 2024). By law, an officer's hand signal overrides a traffic light (Road Transport Act 1987, s. 79). On the MyAV route itself, residents call the junction of Persiaran Rimba Permai and Persiaran Bestari *simpang maut*, the "death junction"; the council posted officers there to direct traffic while signals were installed (Utusan Malaysia, 2025; Malaysia Tribune, 2025). The same road was the worst-hit area in a 2019 flash flood (The Sun, 2019).

This study asks four questions:

1. **RQ1.** How does remote assistance work in current practice, and where does it fail?
2. **RQ2.** Is there precedent for roadside views supporting remote assistance, and would they be new in Malaysia?
3. **RQ3.** What network capacity would such views need, and can Malaysian networks provide it at busy moments?
4. **RQ4.** What safety, privacy and cost rules should a Malaysian deployment follow?

It contributes a synthesis of public evidence on these questions; a design, MATA KEDUA ("second eye" in Malay), that follows from it; a roadside unit costed from public list prices; an interactive prototype; and a measurement protocol for the first phase of a field trial.

2. Background and related work

2.1 Remote assistance is not remote driving

Industry and standards separate remote monitoring, remote assistance and remote driving (Kamtam et al., 2024). In assistance, the vehicle stays in control. Waymo describes a vehicle that asks a question; a human can mark a lane closed or propose a path "for the vehicle to consider", while the car's own system keeps driving (Waymo, 2024). In February 2026 the company stated that its service "does not rely on remote drivers", that about 70 assistance agents were on duty for a fleet of about 3,000 vehicles (roughly 1:43, our calculation), and that median one-way latency to its agents was about 150 ms from US centres and 250 ms from overseas centres (Waymo, 2026). Pony.ai reported a remote-assistant-to-vehicle ratio of 1:20 in May 2025 (Pony.ai, 2025). Zoox describes staff placing waypoints rather than controlling the vehicle (Zoox, 2024; search extract). Remote driving does exist, in Baidu's Apollo Go service (Stanford CIS, 2025; search extract) and, at low speed, at Tesla (Markey, 2026), but it is the exception.

Two jurisdictions have written "confirm, never steer" into law. In Germany, a vehicle proposes a manoeuvre, a human technical supervisor approves it, and the vehicle must refuse an approved manoeuvre that would endanger people (StVG §§ 1e–1f). In Japan, Level 4 services need a remote supervisor who follows police officers' instructions and records video, and anyone operating the steering or brakes counts as driving, which excludes remote driving from Level 4 (National Police Agency, n.d.; search extract).

2.2 Where assistance fails

Two recent cases show the weak points. During a power outage in San Francisco on 20 December 2025, Waymo's cars treated dark signals as four-way stops and crossed more than 7,000 of them, but the outage "created a concentrated spike" in confirmation requests; the backlog delayed responses and added to congestion (Waymo, 2025). In Austin on 12 January 2026, a Waymo stopped and asked whether a school bus had active signals; a remote agent answered "No", and the car passed the bus with its stop arm out (NTSB, 2026a). The lessons are direct: assist capacity must be planned for predictable peaks; a mistaken human answer must fail safe; and the human needs a better view than the one that confused the vehicle.

2.3 Roadside support for remote monitors

Roadside cameras already feed remote monitors in Japan. At Eiheiji, the remote monitoring room of a Level 4 service shows cameras installed at the stops on a dedicated infrastructure-camera monitor (RoAD to the L4, 2023). In Komae, Tokyo, cameras and sensors at intersections and a station roundabout send data over local 5G to the remote monitor, to support the bus's decision to enter (Mitsubishi Research Institute, 2024). Japan's FY2024 programme ran further sites where smart poles report pedestrians and cyclists in blind spots to remote rooms (Mitsubishi Research Institute, 2025; search extract). We found no case of roadside views at school gates or flood-prone dips feeding on-request assistance for a delivery van, though absence of evidence is not proof.

2.4 Edge-case libraries

Scenario libraries exist: Japan's SAKURA database, the UK's Safety Pool and China's CICV anomaly dataset (search extracts). Malaysia's published trial guideline asks only for monthly reports of near-misses,

safety-officer take-overs and complaints (Futurise, 2024). A library of Malaysian assists, labelled by hazard type, does not appear to exist.

2.5 The Malaysian setting

MyAV is a Ministry of Transport initiative run with Futurise under the National Regulatory Sandbox (Futurise, n.d.). The AV Trial Guideline v6.0 requires PDPA compliance and operational data loggers and cameras, but says nothing about remote operation centres (Futurise, 2024). No AV-specific law exists yet; in August 2026 the Digital Minister said legal changes were being considered and that camera data can help reconstruct events before an accident (The Star, 2026b). Cyberjaya has hosted 5G remote monitoring of an autonomous bus before, in a 2024 demonstration by DNB, Ericsson and eMooVit (Digital Nasional Berhad, 2024), and in 2016 the traffic lights on Persiaran Multimedia were fitted with LTE-connected controllers running analytic video cameras (Cyberview, n.d.). Remote monitoring over 5G is therefore not new in Cyberjaya. What public descriptions of the ROC do not include is a view from the roadside.

3. Method

3.1 Desk review

Between 27 and 29 October 2026 we searched English and Malay news, operator and regulator documents, standards, and academic work, with some Japanese and Chinese primary sources, using web-search and page-retrieval tools. We cite about 70 sources; about 35 were read in full. Sources read only as search extracts are marked "search extract" in the text, and figures resting on a single report are marked as such. Key figures were checked against their original pages. An independent mock-judge review of an earlier draft led to corrections, recorded in the project files; for example, a two-vehicle result in one network study concerned LiDAR uploads, not camera views, and is not used here.

3.2 Design and prototype

Findings were turned into design rules (Section 5). We built an interactive web prototype that simulates one 38-second assist at a school gate, plus three side cases (a false alarm, a request the van resolves itself, and a frozen feed), with three views: the operator console, the guard's phone and the engineers' library. All data in it are simulated and labelled so. The prototype was reviewed twice by independent design reviewers. The roadside unit was costed from public list prices in October 2026; US-dollar prices were converted at Bank Negara Malaysia's reference rate of RM 4.0818 per US dollar on 28 October 2026.

3.3 Estimates

One figure is our own arithmetic, labelled as such where it appears: the number of 8 Mbps camera views that fit within published median upload speeds.

3.4 Data we could not obtain

No full public map of the 22 km route exists; which schools and MRT stations it passes, which network the ROC uses, and whether the van can accept a short path from an operator are not public. The team collected no primary data: no drive test, no counts and no survey. These become Phase 1 work (Section 7).

4. Findings

4.1 RQ1: practice converges on advice with a fail-safe default

Across Waymo, Zoox and Pony.ai, and in German and Japanese law, the human advises and the vehicle drives (Section 2.1). The failures on record are failures of view and capacity: a wrong "No" about a school bus, and a backlog of requests in a blackout (Section 2.2). For design, this means three things: questions should be framed so that a mistaken answer leaves the vehicle stopped; the person answering should see more than the vehicle saw; and staffing should follow predictable peaks.

4.2 RQ2: new to Malaysia, not to the world

Roadside views for remote monitors run in Japan (Section 2.3), which is evidence that the mechanism works. We found no Malaysian precedent and no case aimed at school gates or flood dips inside an on-request assist. The defensible novelty is therefore local: on-request roadside views at Malaysian hotspot types, a Malaysian edge-case library, and a measured upload map of the MyAV route.

4.3 RQ3: the constraint is upload capacity at busy moments

Latency is not the binding constraint. Assistance tolerates more delay than remote driving. The 5G Automotive Association puts indirect control at 10 km/h at about 300 ms end to end (5GAA, 2021), and Waymo's measured one-way latency of 150–250 ms supports a working service (Waymo, 2026). Zelostech states that the ROC is not a remote-driving system (Zelostech, 2026).

Upload capacity is. One 1080p camera needed 8 Mbps of upload in Ericsson's 2017 LTE test (Ericsson, 2017); H.265 video takes 4–10 Mbps per camera (DriveU, 2021); the 5GAA assumes about 8 Mbps per HD camera and up to 32 Mbps for four (5GAA, 2021). Four views therefore need roughly 16–32 Mbps, plus the roadside view. Table 1 sets this against Malaysian measurements.

Table 1. Measured mobile upload speeds in Malaysia

Source and period	5G upload	All networks	Note
Opensignal, Apr–Jun 2026	CelcomDigi 32.9 · Maxis 32.3 · Unifi 32.2 · U Mobile 31.2 · Yes 25.9 Mbps	8.5–17.1 Mbps	Averages; Selangor 5G 31.7–32.4 Mbps
Ookla Speedtest, Jan–Jun 2026	—	Putrajaya 38.96 · Selangor 21.86 Mbps	Medians, all networks
Opensignal, 2022 (4G era)	—	2.4–7.1 Mbps	What 4G alone delivered

Sources: Opensignal (2026a, 2022); Ookla (2026).

On our arithmetic, four 8 Mbps views fit within Putrajaya's median upload ($38.96 \div 8 \approx 4.9$) and two within Selangor's ($21.86 \div 8 \approx 2.7$). This uses published medians, not measurements of the route, and no Cyberjaya-specific measurements exist.

Consistency matters as much as averages. In a study on a commercial 5G Standalone network in Minneapolis, a single camera stream compressed with H.265 missed a 100 ms deadline on 0.487% of frames, while three merged cameras missed it on 3.64% (WebRTC) to 13% (RTSP) of frames (Fezeu et al.,

2025). Those deadlines are set for remote driving; for assistance they mark the direction of the problem, not its size. At a school gate at 7:15 am, the van's uplink shares the cell with every parent's phone.

Priority exists; guaranteed slices do not yet. Priority over standard users can be bought on 4G and 5G today (Maxis Business, n.d.; U Mobile, 2026), but it is best-effort. A network slice with a guaranteed uplink requires 5G Standalone and, in Malaysia, would require a trial: we found slicing only in trials and demonstrations. Consumer 5G Standalone is offered by U Mobile (since August 2025) and Yes (since 2025) (Opensignal, 2026b; YTL Communications, 2025).

What 4G can do. On an uncongested LTE cell, one compressed view reaches a remote operator fast enough for a yes-or-no decision (Ericsson, 2017). That suits one van on a quiet road. It does not suit four views and a roadside view at a busy gate.

4.4 RQ4: Malaysian constraints on a deployment

Hazards on or near the route. School zones see an average of 67 deaths a year among all road users (WHO, 2024); a national 30 km/h school-zone limit is approved and being gazetted (The Star, 2026c); the Road Safety Council launched a Safe School Zone pilot at 14 schools in 2026 (NST, 2026b), and at the first site 76.6% of surveyed pupils said they felt unsafe on the surrounding roads (Berita Harian, 2026; single source). Parents report double-parking "in the middle of the road" at school gates (NST, 2024). About two-thirds of Malaysia's 6,581 road deaths in 2025 were motorcyclists or pillion riders (The Star, 2026a), and researchers observed 87% of motorcycles moving through queues at red lights at Klang Valley junctions (Paiman et al., 2020; search extract). Officers' hand signals override signals by law (Road Transport Act 1987, s. 79). Persiaran Rimba Permai has both an officer-controlled junction and a flood history (Section 1).

Privacy. The 2024 amendment to the Personal Data Protection Act makes biometric data sensitive personal data (Act A1727, 2024), so face recognition at a school gate would process sensitive data. The Commissioner's consultation on CCTV proposes cameras only where they serve a specific purpose, prominent notices, no audio unless justified, restricted access and regular deletion (JPDP, 2025).

Public trust. About six in ten Malaysians say they would feel safe riding in a self-driving car, and 61% are concerned about road safety in their area (Ipsos, 2026). Trust is neither absent nor assured.

5. Design: MATA KEDUA

5.1 Overview

MATA KEDUA adds three things to the control room ALS already runs (Figure 1). First, when the van asks for help, the operator receives four camera views and what the van's sensors detect over a prioritised 5G link. Second, at registered hotspots (a school gate at drop-off, a junction with officers directing traffic, a flood-prone dip), a roadside camera's view opens when a van there asks for help or approaches during the hotspot's busy hours. Third, every assist is labelled in a Malaysian edge-case library that ALS can retrain the van with and that regulators receive as anonymised summaries.

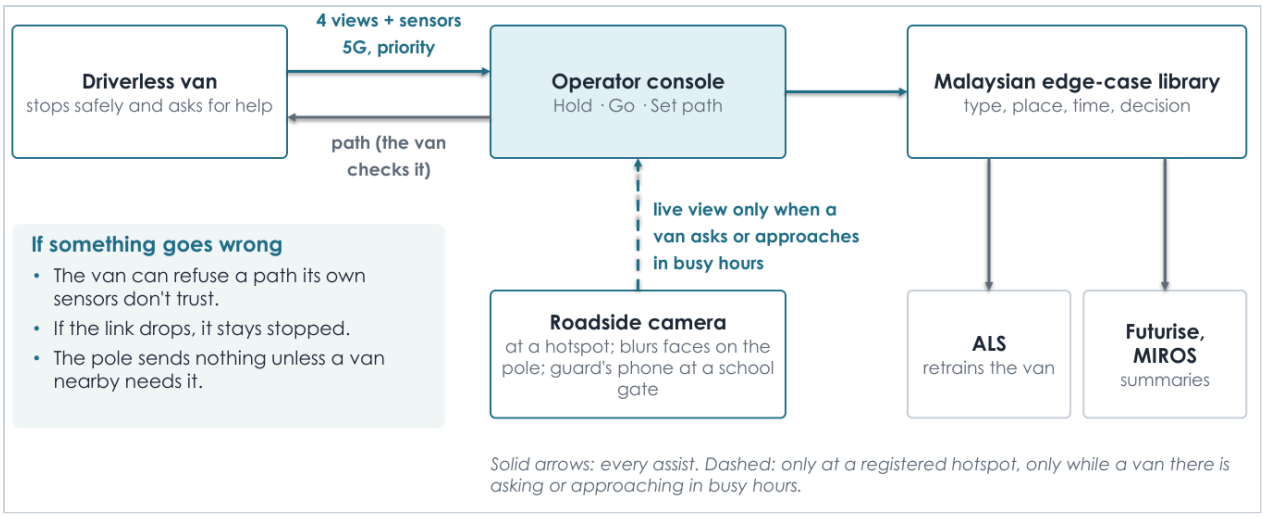


Figure 1. How MATA KEDUA connects. Solid arrows run on every assist; the dashed roadside link opens only at a registered hotspot, only while a van there is asking or approaching in busy hours.

5.2 Interaction and safety rules

The operator answers with one of three actions: **Hold**, **Go** or **Set path**. The operator never steers. A path is a short message the van checks with its own sensors and can refuse, following the German supervisor rule (Section 2.1). The design adds an interlock: **Go and Set path stay locked while the roadside view shows people in the path, while the guard has not answered, or while the roadside feed is lost**, and the reason is shown beside the locked buttons (Figure 2). Hold is always available and always in the same place. If video freezes or the views disagree, the answer stays Hold and ALS's ground team is called. At a school gate (and only in Phase 3), the guard directing traffic can send "clear" with a deliberate one-second press, or "not yet"; it is one more signal, never sufficient alone. Staffing follows the hotspot map, with extra cover at 7:00–7:45 am and in monsoon storms, in response to the San Francisco backlog.

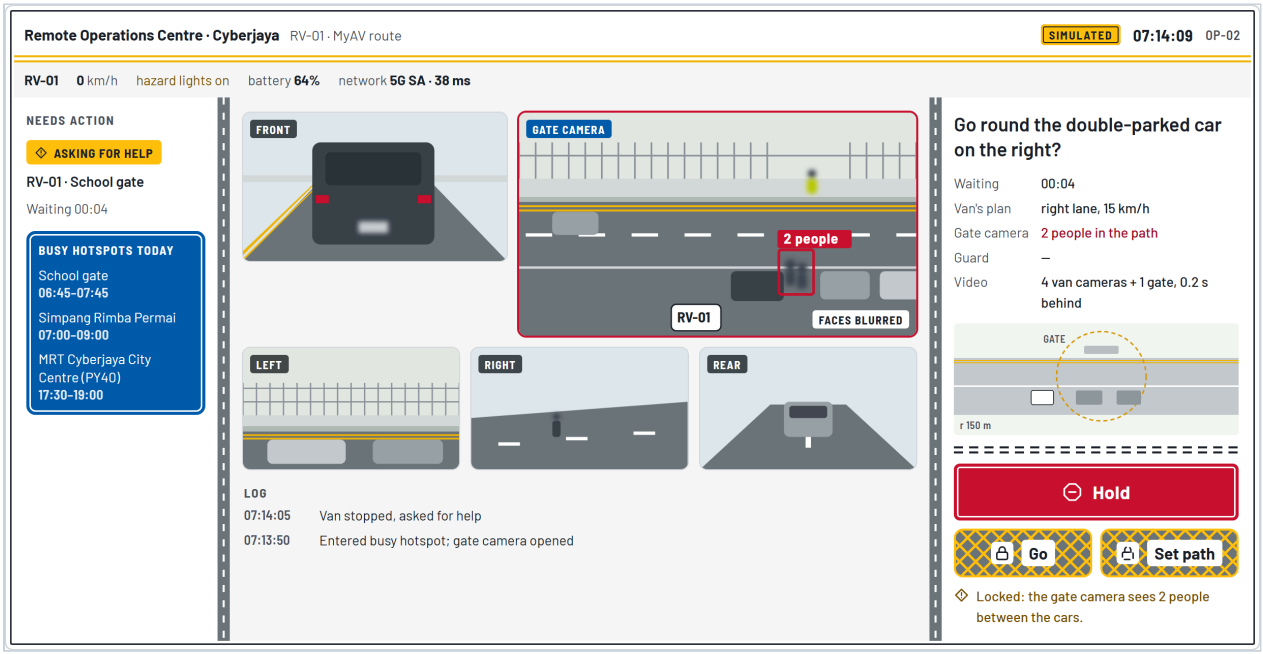


Figure 2. The prototype's operator console at 07:14:05 (simulated). The gate camera sees two people between the cars, so Go and Set path are locked under a yellow box-junction hatch, with the reason written below.

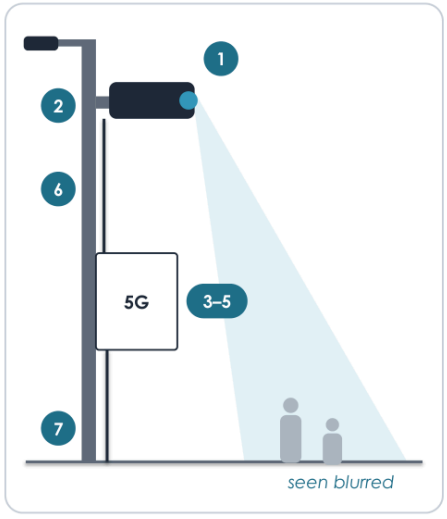
5.3 Privacy by design

Faces and number plates are blurred on the camera before video leaves it: Axis Live Privacy Shield runs on the camera, comes preinstalled, and can mask people, faces and number plates (Axis Communications, n.d.-a). The camera can also send an unblurred stream; the design keeps it off. There is no face recognition and no audio. The view is live only while a van within 150 m is asking for help or approaching in the hotspot's registered busy hours. One blurred 30-second clip is kept per logged assist and deleted after 30 days unless the police or Futurise ask for it. Every camera has signs in Bahasa Melayu and English, and school-gate cameras come only in Phase 3, with the agreement of the school and its parent-teacher association.

Two limits shape placement. The AI masking runs at up to 10 frames per second (Axis Communications, n.d.-b), and where the image has fewer than 100 pixels per metre a person "may not be completely masked" (Axis Communications, n.d.-c). Each camera is therefore placed to cover its crossing at close range.

5.4 The roadside unit and its cost

From Phase 2, each hotspot gets one unit on an existing street-light pole, with nothing installed on the van (Figure 3). The parts come to about RM 7,800 at October 2026 list prices: an AXIS P1465-LE camera (US$839, about RM 3,430), a pole mount (about RM 430), a Teltonika RUTX50 5G router (RM 2,688 at a Malaysian reseller), a 24 V supply rated to 70 °C (RM 215), an IP66 plastic box that lets the 5G signal through (RM 27), an allowance of RM 300 for cables and surge protection, and about 10% contingency. Installation and the mains connection would be quoted by the pole's owner. One supply powers both devices: the camera's PoE Class 3 ceiling of 12.95 W plus the router's maximum of under 16.5 W is about half the supply's 60 W rating (our calculation). Running cost is about RM 200 a month for a business 5G line with priority (Maxis Business, n.d.). Phase 1 needs no roadside unit: a team phone streams recorded footage in its place.



#	Part	RM
1	Camera that blurs faces and plates itself (AXIS P1465-LE)	3,430
2	Stainless steel pole mount (AXIS T91B47)	430
3	5G router, two SIMs, SIRIM-approved (Teltonika RUTX50)	2,690
4	24 V supply for both, rated to 70 °C (Mean Well HDR-60-24)	220
5	IP66 plastic box, so the 5G signal gets through	30
6	Cables, glands, surge protector (our allowance)	300
7	*Installation and mains power from the pole*	*quote*
	Contingency (about 10%)	700
	Total per hotspot, parts only	**7,800**

- **Not in the Phase 1 budget (RM 5,100):** there, a phone stands in for the camera. Phase 2 needs 1–2 units.
- **Running cost:** about RM 200 a month for a business 5G line with priority. Item 7: the pole owner's quote.
- **Limits:** blurring runs at 5–10 frames a second and can miss distant people, so each camera covers its crossing up close. Its unblurred stream stays off.

Figure 3. The roadside unit: a numbered pole drawing and its parts list at October 2026 list prices. Item 7 (installation and mains power) is quoted by the pole owner.

5.5 The prototype

The interactive prototype (https://zghanw.github.io/matakedua/) plays the simulated assist on its own and hands control to the viewer on any click. It demonstrates the interaction logic and the interlock, not network performance or detection accuracy; its camera views are drawings, not footage.

6. Discussion

6.1 What the evidence supports

The mechanism is plausible because it already runs in Japan. The 5G case is specific: the value of 5G here is upload room and the option of prioritisation at the busiest moment, not lower latency. The safety rules map onto the failures on record: the interlock and the Hold default answer the wrong "No"; the hotspot-based staffing answers the request spike; the roadside view answers the view that cannot see past a parked car.

6.2 What it does not show

We have not measured anything on the route. National averages may not hold in a particular cell at 7:15 am, and no Cyberjaya-specific upload data exist. We do not know whether the ROC's network or the van's software can accept the additions, which depends on ALS. We have not measured any safety benefit; the impact measures in the proposal (assist time, lane-blocking minutes, holds for hidden people) are targets for Phases 2 and 3. Operator workload with a second view is untested, and on-camera masking has known limits.

6.3 Threats to validity

Opensignal reports averages and Ookla reports medians, so the two are not directly comparable. Several sources were read only as search extracts and are marked. Some Malaysian figures rest on single reports, such as the 76.6% survey result. Prices are list prices, not quotes, and imported hardware may cost more in Malaysia once duties and tax are added.

6.4 Ethics

The design involves cameras near children. It avoids face recognition, blurs on the device, records only per assist with short retention, and defers school-gate cameras to a phase that requires the school's consent. This paper names no victims and uses no footage.

7. Conclusion and future work

Remote assistance asks a person to decide from the vehicle's point of view. At Malaysian hotspots, that view can be the problem. The evidence suggests that a roadside view, opened only when needed and paired with an interlock, is a feasible addition, and that 5G matters for carrying it at the busiest moments. The claims need field data, so we propose a first phase that needs no one's permission (November 2026 to January 2027, about RM 5,100):

- **Drive test.** A laptop and 5G router in a car stream four recorded camera views on 4G, 5G and a priority line at 20 or more stops, three times of day; a phone at each hotspot streams a fifth recorded view. No one is filmed.

- **Hotspot counts** at five sites, including monsoon storms, and a resident survey.
- **Go or no-go.** Proceed if 5G carries four views plus a roadside view at most hotspots at peak and at least three hotspots show recurring edge cases. If 4G proves enough, ALS saves money and we publish that result.

Later phases would offer ALS, Sidec, Cyberview and Futurise a shadow trial at one or two non-school hotspots, with the roadside view on a separate screen, and only then a first school gate with consent.

Statement on the use of AI

We used AI tools to search for and summarise sources, to draft and tighten wording, to review drafts as a mock judge and designer, and to help build the interactive prototype. We checked key figures against their original pages. The concept, the design decisions and the conclusions are our own.

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